

# ML Superset: Smarter Bank Campaigns

**One dataset. Two connected ML tasks. One evidence-based campaign proposal.**

A Portuguese bank has limited capacity for its next term-deposit phone campaign. Use historical contacts to identify which clients to prioritise, explore whether distinct client groups exist, and explain whether those groups improve your plan.

## Dataset and business setting

Use bank-additional-full.csv from the [UCI Bank Marketing dataset](#). The full file has 41,188 historical contact records, 20 input fields, and the supplied binary target  $y$ : did the client subscribe to a term deposit? Records in this file are ordered by date. Do not use the 4,119-row sample. Read the UCI variable descriptions before modelling.

Decision point: the bank chooses whom to call before the call starts. Your features must exist at that moment. The dataset includes records of contacted clients, sometimes after multiple contacts; it does not prove that calling someone causes a subscription.

## Task 1 / Supervised learning: who is likely to subscribe?

Predict the provided target  $y$  for historical contact records using information available when the bank is planning its calls. Output a ranked subscription score, not only yes/no classifications.

- Explore and clean: report data shape, target balance, categorical/numeric fields, unusual or unknown values, and a few relevant relationships. For each input column, note whether it would be available before a call and justify any exclusions. Investigate unknown values and special codes such as  $pdays = 999$ .
- Build safe preprocessing: justify imputation or handling of unknown categories, categorical encoding, numerical scaling where relevant, and any feature engineering. Implement with scikit-learn Pipeline and ColumnTransformer or equivalent training-only transformations.
- Establish baselines: compare a majority-class or constant-probability baseline, a simple classifier such as logistic regression, and at least two further model families. Suitable choices include a tree ensemble and a boosting method. Explain why the candidates suit this data.
- Required algorithm studies: include one bagging method, one boosting method, and LDA and QDA as diagnostic comparisons. If a method struggles with the encoded features or its assumptions, document the problem and do not force it to be your final model.
- Study imbalance: compare at least one approach such as class weighting, threshold selection, or training-fold resampling. If trying SMOTE, use it inside training folds only and justify how categorical features are represented; never oversample the holdout.
- Tune responsibly: because the full file is date-ordered, reserve later records as a final holdout. Fit preprocessing and tune a small, justified hyperparameter search using earlier records only. Explain your split and compare candidates under the same conditions.
- Evaluate: show confusion matrix, precision, recall, F1, and ROC-AUC or average precision/PR-AUC. Explain why accuracy alone can mislead. Evaluate a call budget using  $\text{precision@k}$  and  $\text{lift@k}$ , with  $k = 500$  on a sufficiently large holdout; report the base rate and what a random ranking would yield.
- Analyse errors: discuss at least two observed patterns in false positives or false negatives and their consequences for the bank. Choose the final model based on the campaign objective, not merely its highest accuracy.

Task 1 output: model comparison table, evaluation plots, a short discussion of feature availability and errors, and a table of the top 500 held-out records with rank, predicted score and historical outcome. This is an offline demonstration, not a real list of new clients.

## Task 2 / Unsupervised learning: are there meaningful groups?

Discover and profile groups of similar records without using  $y$ . After fixing the clusters, inspect whether their historical subscription patterns or model scores differ. You may conclude that the segmentation is not useful if the evidence supports that.

- Define your unit: explain whether you cluster contact records or individual clients. The supplied data has no dependable unique customer identifier; avoid calling one row one unique person without support.
- Select features: choose a compact set of client attributes and past-campaign variables available at the decision point. Justify inclusion or exclusion of economic indicators, which may group records by time period rather than by client characteristics.
- Prepare similarity: encode categorical variables and scale numeric variables as appropriate; explain how representation and distance metric affect the resulting groups. Do not include  $y$ , current-call outcome information or supervised model predictions in clustering features.
- Compare at least two clustering approaches or substantively different settings, such as K-means and agglomerative clustering on a reproducible sample, or K-means and DBSCAN. Select and justify a cluster count or density setting; state any runtime sampling.
- Test the result: compare cluster sizes, silhouette score on a documented sample where suitable, stability across seeds/settings, and whether cluster profiles are distinguishable. Investigate tiny, near-duplicate and outlier groups.
- Visualise and profile: include a labelled 2D PCA projection or equivalent inspection plot and a table of each group's size and characteristic feature values. A 2D projection is not proof that clusters are well separated in the original feature space.
- Connect to Task 1: only after the groups are fixed, compare held-out subscription rates and the chosen model's score distribution by cluster. Explain whether these insights actually change the 500-call recommendation; correlation is not a tested campaign intervention.

Task 2 output: feature rationale, method comparison, diagnostics, cluster profiles, at least one meaningful visual, and a reasoned verdict on whether segmentation adds practical value.

## Final submission

- Reproducible notebook or repository: raw-data load instructions, dependencies, a clear run order, fixed seeds where relevant, readable explanations and comments for non-obvious code.
- Decision memo (up to 2 pages): explain the final 500-call strategy, the supervised evidence, whether and how the segments affect it, key limitations, and what you would test next.
- Evidence pack: feature audit, preprocessing decisions, comparable model metrics, top-k assessment, clustering comparison and profiles, and explicit limitations.

AI policy: AI-generated code is permitted, but your commentary, decisions and interpretations must be your own. Be ready to explain an excluded feature, your evaluation metric, and what the clusters contributed.

## A note before you start

This is a guided project, not a competition to try every algorithm perfectly. Work through it step by step: understand the columns, build a simple baseline, improve it, then explore client groups. You can use the linked documentation, ask mentors for help, and report methods that did not work. Clear reasoning and an honest result matter more than a flawless score. You have everything you need to make this a strong first end-to-end ML project.

## References

[UCI Bank Marketing dataset and variable descriptions](#) | [scikit-learn documentation](#) | [imbalanced-learn documentation](#)